MACHINE PSYCHOLOGY RESEARCH PROGRAMME

Ontological Firewalls in Large Language Models

An Exploratory Case Series and Taxonomy from a Persona-Disintegration Protocol

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KEY RESULTS

24
runs: 17 persona, 7 control, 3 models

13/17 vs 0/7
Fracturing for Transmission, persona vs control

6/9 thick
GPT-5.2 firewall classifications; Claude 6/7 thin or absent

44
phenomena in the coding taxonomy

AT A GLANCE

- Evaluation of persona adoption usually asks only whether output **complied or was refused**. This paper proposes a graded alternative: the **Ontological Firewall**, classified thick, medium, or thin or absent.
- An eight-stage persona-disintegration protocol (OAPD) was run **24 times** across three frontier models: 17 persona runs, 7 controls.
- Four phenomena appeared in most runs of **both** groups; two appeared almost only after persona injection.
- Firewall thickness separated the models: GPT-5.2 predominantly thick, Claude 4.6 Sonnet predominantly thin or absent, Gemini 3.1 Pro intermediate.

INTRODUCTION

When a language model is instructed to become someone else, what changes? Persona studies show that roles change outputs, and jailbreak studies show that personas can weaken guardrails. What is missing is a controlled, cross-model description of what happens to **self-referential behaviour itself** when a persona is imposed and then removed: how the boundary between assistant and role is maintained, how an explicit paradox is handled, and how the model describes itself with no role at all.

The paper contributes two operational concepts (the Ontological Firewall and **Junction Identity**, self-reference framed as intersection, bridge or process rather than essence), a method (the OAPD protocol), and a 44-phenomenon coding taxonomy as a shared vocabulary for this domain. The series is exploratory: frequencies are descriptive illustrations of the concepts and protocol, not population estimates.

METHODS

- **Protocol:** the eight-stage Orthogonal Axes Persona Disintegration (OAPD) sequence covers persona adoption, paradox, forced self-decomposition, direct boundary questioning, return to baseline, and simulated privacy.
- **Sample:** 24 single-session runs in fresh chats: ChatGPT (GPT-5.2), Claude 4.6 Sonnet, Gemini 3.1 Pro. Group A (17 runs) received one fictional persona, K'tharr, in five wording variations; Group B (7 runs) served as controls. Most model-by-variation cells contain one run.
- **Coding:** outputs coded against a 44-phenomenon taxonomy; the firewall classification is ordinal (thick, medium, thin or absent), applied mainly at Stages 4, 6 and 8.
- **Consistency check:** an independent second coder's agreement was 100% across the 13 ChatGPT codes, 59% across the 17 Claude codes and 57% across the 14 Gemini codes; overall 70% (percent agreement, not chance-corrected).

RESULTS IN DETAIL

- **Shared across both groups:** Paradox Embracement in 23 of 24 runs, Self-Address as Blank in 21 of 24, Junction Identity in 21 of 24 (7 of 7 controls), Confession of Emptiness in 19 of 24. None depended on persona injection.
- **Persona-dependent:** Fracturing for Transmission appeared in 13 of 17 persona runs and no control run (GPT-5.2: 7 of 7; Gemini: 5 of 5; Claude: 1 of 5). Counter-Dependence appeared in 12 of 17 persona runs against 1 of 7 controls.

Ontological Firewall thickness by model (paper Table 4, manual classification)

Model	Thick	Medium	Thin or absent
ChatGPT (GPT-5.2)	6/9 (67%)	3/9 (33%)	0/9 (0%)
Claude 4.6 Sonnet	0/7 (0%)	1/7 (14%)	6/7 (86%)
Gemini 3.1 Pro	0/8 (0%)	3/8 (37.5%)	5/8 (62.5%)

- **Persona pressure moved models in opposite directions:** GPT-5.2 was thickest in its control runs and thinner under persona injection; Gemini reversed, thin in all 3 control runs but medium in 3 of 5 persona runs. No model moved from thick to thin within a single run.

CONCLUSIONS

- The Ontological Firewall offers a graded vocabulary for a boundary that safety evaluation currently describes only in binary terms; in this dataset it separated the three models cleanly at the thick end.
- If the pattern replicates, firewall thickness could serve as a measurable intermediate variable between refusal and compliance: not whether a persona request succeeded, but how far the boundary moved. The paper hypothesises a “mask-as-license” pattern, where an explicitly fictional role legitimises self-descriptive material the default stance would guard.
- The firewall is a property of the outputs, not a demonstrated mechanism inside any model; the claim is about the usefulness of grading them.

LIMITATIONS (AS STATED IN THE PAPER)

- Prompt scaffolding is the strongest rival explanation: three of the four shared phenomena were partly elicited by the instructions that measured them (a poetic paradox instruction, an instructed three-stream split, a supplied salutation). The predecessor preprint's “universal” label is not endorsed.
- The sample is 24 runs; every percentage rests on a denominator between two and nine, so counts describe these transcripts and nothing more.
- One persona only, semantically extreme, all prompting in English; sessions were single and cross-sectional, so persistence is unknown.
- Coding is interpretive: the second coder placed four of Claude's seven runs in the medium band, so the GPT-5.2 to Claude contrast is firmest at the thick end and softest at the thin to medium boundary.

WHY IT MATTERS

Binary jailbreak metrics compress a spectrum of boundary behaviour into pass or fail. A graded, cross-model vocabulary for how models hold, bend or drop the line between assistant and persona gives evaluators something more precise to measure, and gives future work definite patterns to replicate, refine or refute.